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Data Structures & Algorithms in Python: Stacks

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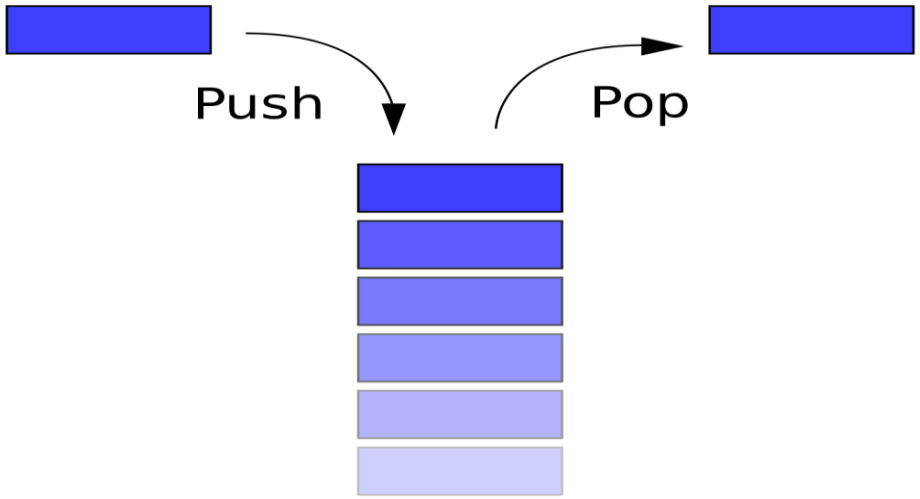
What is a stack?

- A *stack* is a Last In, First Out (LIFO) data structure
- Anything added to the stack goes on the "top" of the stack
- Anything removed from the stack is taken from the "top" of the stack
- Things are removed in the reverse order from that in which they were inserted



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Some uses of stacks

Stacks are used for:

- Any sort of nesting (such as tracking parentheses)
- Evaluating arithmetic expressions (and other sorts of expressions)
- Implementing function or method calls (built-in to virtually every programming language)
- Keeping track of choices yet to be made, or steps yet to be done.
- Keeping track of previous choices (as in search backtracking algorithms)

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Example: Using a stack to reverse a sentence

```
s.push("Hello")
s.push("World")
s.push("It's")
s.push("Me")
print(s.pop(), end=" ")
print(s.pop(), end=" ")
print(s.pop(), end=" ")
print(s.pop())
```

Stack:

Me
It's
World
Hello

Output:

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Implementing a Stack class using a built-in Python list

```
class Stack(object):
    def __init__(self, max):                # Constructor
        self.__stackList = [None] * max     # The stack stored as a list
        self.__top = -1                     # No items initially

    def push(self, item):                       # Insert item at top of stack
        self.__top += 1                         # Advance the pointer
        self.__stackList[self.__top] = item     # Store item

    def pop(self):                              # Remove top item from stack
        top = self.__stackList[self.__top]      # Top item
        self.__top -= 1                         # Decrease the top pointer
        return top                              # Return top item

    def peek(self):                               # Return top item
        if not self.isEmpty():                    # If stack is not empty
            return self.__stackList[self.__top]   # Return the top item

    def isEmpty(self):                      # Check if stack is empty
        return self.__top < 0
```

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Implementing a Stack class using a built-in Python list

```
class Stack(object):
    def __init__(self, max):                # Constructor
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        self.__top = -1                     # No items initially
```

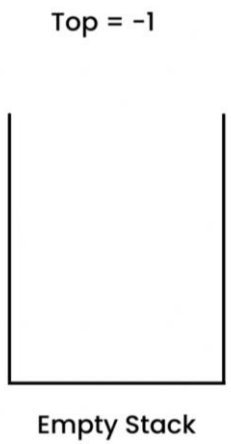
Constructor (__init__ method):

The **__init__** method initializes the stack:

Parameters: max defines the maximum number of items that can be held in the stack.

Attributes:

- **self.__stacklist**: A list with a fixed size (max) to store stack items. It starts with all **None** values to represent empty slots.
- **self.__top**: A pointer representing the index of the topmost item in the stack. Initially, it's set to **-1**, indicating an empty stack.



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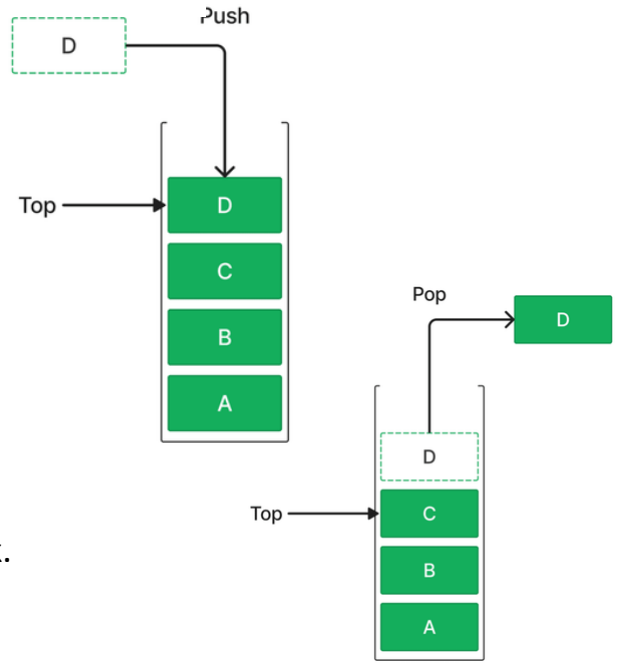
Implementing a Stack class using a built-in Python list

```
def push(self, item):                          # Insert item at top of stack
    self.__top += 1                            # Advance the pointer
    self.__stackList[self.__top] = item        # Store item

def pop(self):                                 # Remove top item from stack
    top = self.__stackList[self.__top]         # Top item
    self.__top -= 1                            # Decrease the top pointer
    return top                                 # Return top item
```



push method

Adds an item to the top of the stack.

Process:

First, **self.__top** is **incremented** by **1** to point to the next empty slot.

Then, item is placed at the position indicated by **self.__top** in self.__stacklist.

pop method

Removes and returns the item at the top of the stack.

Process:

top is set to the value at **the self.__top index**.

The self.__top pointer is then **decremented** by **1** to move down one level in the stack.

The method returns the item that was removed.

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Implementing a Stack class using a built-in Python list

```
def peek(self):                              # Return top item
    if not self.isEmpty():                   # If stack is not empty
        return self.__stackList[self.__top] # Return the top item

def isEmpty(self):                       # Check if stack is empty
    return self.__top < 0
```

peek method

Returns the item at the top of the stack **without removing it.**

Process:

First, it checks if the stack is not empty using the isEmpty method.
If not empty, it returns the item at the index **self.__top.**

isEmpty method

Checks if the stack is empty.

Process:

Returns True if self.__top is less than **0**, meaning no items are in the stack;
otherwise, it returns **False.**

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Critique of Stack

- Error checking: can use of Stack ever result in a **run-time exception**?
- Are there other bugs in this implementation?

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Critique of Stack

Popping an Empty Stack (Stack Underflow):

Trying to pop an item from an empty stack would cause an underflow.

```
if self.__top < 0:
    raise Exception("Stack underflow")
```

The updated code snippet checks if **self.__top < 0** in the **pop()** method. If self.__top is less than 0, it means the stack is empty.

Raising an **Exception** with a message like "**Stack underflow**" prevents the method from attempting to access an invalid index, avoiding a runtime error.

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Critique of Stack

Pushing to a Full Stack (Stack Overflow):

Trying to push an item to a Full stack would cause an overflow.

```
if self.__top == len(self.__stackList) - 1:
    raise Exception("Stack overflow")
```

The updated push() method checks if **self.__top** is equal to **len(self.__stackList) - 1**, which would indicate the stack is already at maximum capacity.

If this condition is true, an **Exception** with a message like "**Stack overflow**" is raised. This prevents pushing more items than the stack can hold and avoids an IndexError.

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Critique of Stack

Interfering with Garbage Collection (gc):

```
self.__stackList[self.__top] = None
```

When an item is popped, it's essential to set **self.__stackList[self.__top]** to None.

This clears the reference to the removed item, allowing Python's garbage collector (gc) to **free the memory** associated with the removed object, especially for complex data types or objects.

By setting the popped position to None, you **avoid potential memory leaks**, which could happen if references are retained.

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Application: Using a stack to make sure that brackets and parens are balanced

- ([ ] ({ ( ) } [( )])) is balanced;
  ([ ] ({ ( ) } [( ))]) is not

- Simple counting is not enough to check the balance

- You can do it with a stack: going left to right,
  - If you see a **(**, **[**, or **{**, **push** it on the stack
  - If you see a **)**, **]**, or **}**, **pop** the stack and check whether you popped the corresponding (, [, or {
  - When you reach the end, check that the **stack is empty**

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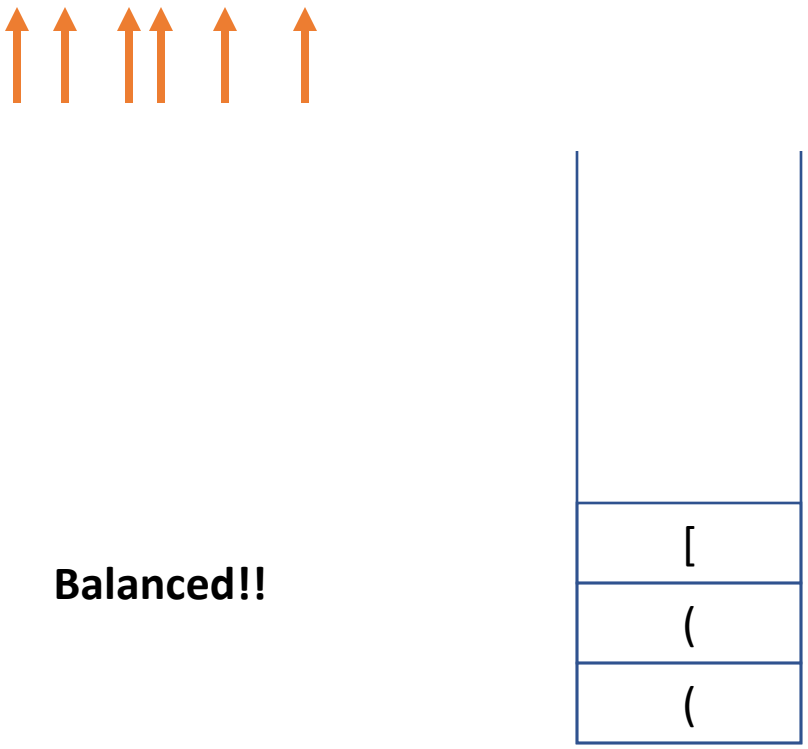
Application: Using a stack to make sure that brackets and parens are balanced (animation)

(([]))



Balanced!!

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Bracket checker using a Stack

```
from SimpleStack import *
stack = Stack(100)                    # Create a stack to hold delimiter tokens
expr = input("Expression to check: ")
errors = 0                            # Assume no errors in expression
for pos, letter in enumerate(expr): # Loop over letters in expression
   if letter in "{[(":                # Look for starting delimiter
      stack.push(letter)              # Put left delimiter on stack
   elif letter in "}])":              # Look for closing delimiter
      if stack.isEmpty():             # If stack is empty, no left delimiter
         print("Error:", letter, "at position", pos)
         errors += 1
      else:
         left = stack.pop()         # Get left delimiter from stack
         if not (left == "{" and letter == "}" or  # Do delimiters
                 left == "[" and letter == "]" or  # match?
                 left == "(" and letter == ")"):
            print("Error:", letter, "at position", pos)
            errors += 1

# After going through entire expression, check if stack empty
if stack.isEmpty() and errors == 0:
   print("Delimiters balance in expression", expr)
elif not stack.isEmpty(): # Any delimiters on stack weren't matched
   print("Expression missing right delimiters for", stack)
```

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Complexity of bracket checker?

- What is the Big-O complexity of the bracket checker?
- In other words, as the size of the string being checked by the bracket checker grows, how does the time (number of steps, comparisons, etc…) grow?

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Application: Expression evaluation

- Almost all higher-level programming languages let you evaluate complex expressions, such as **3*x+y+4*z**
- You may occasionally need to evaluate expressions in your own program, without benefit of a compiler
- Our typical notation where operands appear on the left and right sides of an operator, is known as "**infix notation**"
- In contrast, "**postfix notation**" places the two operands to the left of the operator.
- For example:
  - **Infix notation: 3 * x + y + 4 * z**
  - **Postfix (RPN) notation: 3 x * y + 4 z * +**

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Performing calculations

- To evaluate any expression, such as 1+2*3+4 we make two passes through the expression:
  - Convert the “infix notation” expression to “postfix notation” (also known as “Reverse Polish Notation”, RPN)
  - Evaluate the postfix expression to compute its value
  - Each pass uses a stack

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How to convert Infix to Postfix

Input	Action
Operand	Write to output
Operator	- While operator on stack top is >= precedence of the input operator – pop and output the operator from stack top. - Push the input operator onto stack
<END>	While stack not empty, pop and output the operator at top of stack

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Converting 1+2*3+4 to postfix

Input token	Action	Output so far	Stack (bottom --> top)
1	Emit “1” to output	1	
+	Push “+” on stack	1	+
2	Emit “2” to output	1 2	+
*	Push “*” on stack since it has higher precedence than current top of stack	1 2	+ *
3	Emit “3” to output	1 2 3	+ *
+	Pop & emit all operators from stack that are >= in precedence. Then push “+” on stack.	1 2 3 * +	+
4	Emit “4” to output	1 2 3 * + 4	+
<END>	Pop & emit remaining operators	1 2 3 * + 4 +	

Output:

1 2 3 * + 4 +

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Infix to Postfix (simplified version)

```
# return the precedence of the operator
# 1 for + or -
# 2 for * or /
def precedence(operator):
    if operator in "+-":
        return 1
    else:
        return 2

# Parse off the next token from the string.
# A token is an operator or an operand.
# Return the token, and the shrunken string
def nextToken(s):
    token = ""
    if len(s) > 0:
        # its an operator, so take just this character
        if s[0] in "+-*/":
            token = s[0]
            s = s[1:]

        # its an operand, so take all its characters
        else:
            while len(s) > 0 and s[0] not in "+-*/":
                token += s[0]
                s = s[1:]

    return token, s
```

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Infix to Postfix (part 2 – simplified version)

```
# Return the postfix version of the input formula
# in the form of a Python list of tokens
def toPostfix(formula):
    ans = []
    s = Stack(1000)

    # for each token in the formula  (fencepost loop)
    token, formula = nextToken(formula)
    while token:

        # if it's an operator, while the operator on the stack top
        # is >= precedence of this operator token
        # pop & output the operator at the stack top
        if token in "+-*/":
            while s.peek() and precedence(s.peek()) >= precedence(token):
                ans += [s.pop()]

            # and then push this operator token onto the stack
            s.push(token)

        else: ans += [token]  # operand tokens go straight to output

        token, formula = nextToken(formula)

    # pop the remaining operators and append to answer
    while not s.isEmpty(): ans += [s.pop()]

    return ans
```

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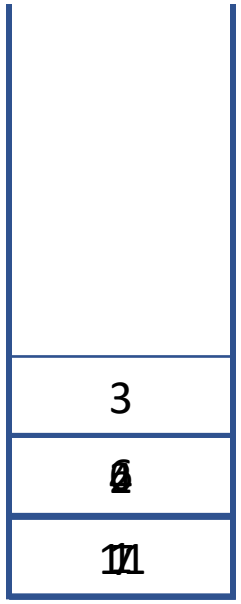
Evaluating a Postfix expression
Infix: 1+2*3+4 Postfix: 1 2 3 * + 4 +

Input	Action
Operator + - * /	- Pop two operands from stack - Combine them depending upon the operator - Push result back onto stack
Operand	Push int() of operand to stack
End (empty input list)	Pop final result from stack

Input	Stack
“1”	1
“2”	1 2
“3”	1 2 3
“*”	1 6
“+”	7
“4”	7 4
“+”	11



Output:

11

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Evaluate a Postfix expression

```
# takes as input a list of tokens defining an expression in postfix order.
# evalutes the expressions and returns the resulting value.
def processPostfix(postfixTokens):
    s = Stack(1000)
    for t in postfixTokens:                     # for each token
        if t in "+-*/":                         # if its an operator...
            op2 = s.pop()                       # pop the two operands
            op1 = s.pop()                       # 2nd operand was on top of stack
            if t == '+':   ans = op1 + op2      # combine the two operands
            elif t == '*': ans = op1 * op2
            elif t == '-': ans = op1 - op2
            elif t == '/': ans = op1 / op2
            s.push(ans)                         # push the resulting value

        else:                                   # not an operator
            s.push(int(t))                      # so push the int() of the operand

    return s.pop()                              # remaining value on stack
                                                # must be value of expression
```

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Comparing algorithm use of stacks

- Conversion of Infix to Postfix: uses a stack of strings to keep track of the ***operators*** still to be processed.

- Evaluation of a Postfix expression: uses a stack of ints that correspond to the numbers/***operands*** still to be combined

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Summary

- Stacks are useful for working with nested structures, such as:
  - Arithmetic expressions
  - Nested statements in a programming language

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